HAZOP Guidewords, Deviations and Potential Causes

For use in HAZOP studies in the chemical / petrochemical industry and the steel & metals industry

This reference expands the continuous-process “HAZOP Guidewords Draft” (guidewords for each line, plus overview guidewords). It is a prompt list, not a substitute for team judgement: the team should still ask “what else could cause this?” at every node.

How to read the table: the Deviation is the guideword applied to the Parameter. Causes without a tag apply to both industries. **[C]** = mainly chemical / petrochemical. **[S]** = mainly steel & metals (iron-making, DRI, EAF / BOF, ladle metallurgy, casting, rolling, finishing, non-ferrous).

Structure: Part A – process parameters for each line / node; Part B – steel & metals specific parameters; Part C – auxiliary guidewords for each line; Part D – overview guidewords for the whole study.

1. Standard HAZOP Guidewords (IEC 61882)

Guideword	Meaning	Typical deviations (chemical / steel & metals)
NO / NOT / NONE	Complete negation of the design intent	No flow, no agitation, no cooling water, no inerting
MORE (HIGH)	Quantitative increase	More flow, high pressure, high temperature, high level, more reaction
LESS (LOW)	Quantitative decrease	Less flow, low pressure, low temperature, low level, low concentration
AS WELL AS	Qualitative increase – something extra is present	Water in molten metal, air in fuel gas, extra phase, impurities
PART OF	Qualitative decrease – only part of the intent is achieved	One component missing, wrong ratio, incomplete reduction
REVERSE	Logical opposite of the intent	Reverse flow, backflow into utility headers, reverse rotation
OTHER THAN	Complete substitution – something entirely different happens	Wrong material, wrong destination, abnormal operation, maintenance mode
EARLY / LATE	Relative to clock time	Tapping late, purge ended early, sample result late
BEFORE / AFTER	Relative to order or sequence	Oxygen on before lance in position, charging before furnace is drained of water
FASTER / SLOWER	Rate of a step or change	Casting speed too fast, heat-up rate too fast for refractory

2. Parameter – Deviation – Potential Causes

Parameter	Deviation	Potential Causes
Part A – Process Parameters for Each Line / Node (Chemical and Steel & Metals) *Apply to every line, vessel or node in turn (continuous process). Causes from the original draft are retained and expanded; industry-specific causes are tagged.*		
Flow	**No flow**	• Empty source vessel, feed tank, bin or gas holder • Wrong routing / incorrect valve line-up • Blockage, gas lock or vapour lock • Slip plate / spectacle blind left in or incorrectly installed • Non-return valve fitted the wrong way round • Burst pipe or large leak upstream • Equipment failure – control valve fails closed, isolation valve, pump or compressor trip, vessel failure • Incorrect pressure differential (downstream pressure higher than upstream) • Isolation in error (wrong valve closed, LOTO on wrong equipment) • Power failure; instrument air failure (valves go to fail position) • Control system / DCS / PLC failure; spurious trip • Line frozen or solidified (loss of heat tracing or jacket heating) • [C] Polymerisation, crystallisation or hydrate formation in the line • [S] Molten metal or slag frozen in launder, runner, tap hole or nozzle (skull formation) • [S] Blocked tuyere, lance tip, burner or porous plug • [S] Bridging / hang-up of burden, DRI, pellets or scrap in bins, chutes or shaft furnace • [S] Conveyor, feeder or vibratory feeder stopped (belt tear, drive trip, chute blockage)
Flow	**Reverse flow**	• Defective or passing non-return valve • Siphon effect • Incorrect differential pressure • Two-way flow in common headers • Poor isolation / passing valves • Emergency venting or depressurisation • Incorrect operation (valve opened out of sequence) • Pump failure or pump running in reverse (wrong rotation after motor rewiring) • [C] Process fluid backflow into utility lines (nitrogen, steam, cooling water, potable water) when utility pressure is lower than process • [S] CO-rich process gas (BFG, COG, BOF/LD gas, DRI top gas) backflowing into nitrogen, air or steam headers • [S] Flashback in fuel-gas, oxygen or oxy-fuel lines (flame arrestor missing or failed) • [S] Loss of water seal / seal pot allowing gas back-flow into the plant or atmosphere • [S] Reverse flow of cooling water due to loss of pump with elevated header (draining of cooling circuits)
Flow	**More flow**	• High level in upstream vessel (higher static head)

Parameter	Deviation	Potential Causes
		• Increased pumping capacity (parallel pumps started, larger impeller fitted) • Increased suction pressure / reduced delivery head • Greater fluid density • Exchanger tube leaks (high-pressure side into low-pressure side) • Restriction orifice plates deleted or scale dislodged • Cross-connection of systems • Control faults – control valve fails open, bypass left open, wrong set point • Two-phase flow • Power failure (valves failing open) • [C] Reagent dosing pump over-delivery (wrong stroke setting) • [S] Excessive oxygen flow to lance, burners or tuyeres (flow controller fault, wrong recipe) • [S] Over-feeding of DRI, scrap, fluxes or alloys (feeder speed, weigh-hopper failure) • [S] Excess fuel gas to reheat furnace, ladle pre-heater or stove burners
Flow	**Less flow**	• Low level in upstream vessel • Line restriction; valve partly closed • Filter or strainer blockage • Pipe or vessel leakage • Defective pumps (worn impeller, cavitation, air ingress, seal failure) • Fouling of vessels, valves, restrictors or orifice plates • Density or viscosity problems • Incorrect specification of process fluid • Two-phase flow • [C] Catalyst bed plugging or high pressure drop • [S] Scale, silt or biological fouling in cooling passages of panels, roof, tuyeres, staves, lances or copper moulds • [S] Partly blocked tap hole or nozzle (low tapping rate) • [S] Dust build-up in off-gas ducts, fans, baghouses or venturi scrubbers
Flow	**Misdirected flow (Other than – wrong destination)**	• Cross-connection between systems; wrong hose connected • Road / rail tanker discharged into wrong storage tank • Drain or vent valves left open • Utility station connected to process (nitrogen hose left connected) • [S] Molten metal or slag spilled outside ladle, pot or mould (ladle misaligned, crane positioning error, tilting over-travel) • [S] Slag carry-over into ladle or tundish • [S] Off-gas diverted to the wrong system (flare / stack / gas holder change-over error)
Flow	**As well as (additional material or phase in the flow)**	• Gas or air entrainment in liquid; liquid carry-over in gas • Water in hydrocarbon or solvent

Parameter	Deviation	Potential Causes
		• [C] Unreacted material carried forward to next stage • [S] Water, ice or snow carried into the furnace with wet scrap, DRI or fluxes • [S] Entrained dust or fines in process gas (cyclone or scrubber failure) • [S] Oil, grease or paint on scrap generating fume and fire
Pressure	**More pressure**	• Surge problems (water hammer, fast-closing valves) • Leakage from interconnected high-pressure system • Gas breakthrough (inadequate venting / pressure relief) • Isolation procedures for relief valves defective (relief valve isolated) • Thermal overpressure / flashing / boiling / external fire • Decomposition / gas evolution • Runaway reaction • Positive displacement pumps or compressors against closed outlet • Failed-open pressure control valves (PCVs) • Incorrect operation • Blocked outlet, blocked vent or blocked flare / stack • Thermal expansion of trapped (blocked-in) liquid • [S] Steam generation from water contacting molten metal or slag (water-cooled element leak, wet charge) • [S] Furnace or converter off-gas pressure excursion (loss of draft fan, damper closed, slopping) • [S] Blast-furnace top-pressure surge from burden slip / hanging; hot-blast stove valve fault • [S] Gas holder over-fill or seal failure (BFG, COG, LD gas) • [S] Hydraulic system overpressure (relief valve set wrongly, accumulator)
Pressure	**Less pressure (vacuum)**	• Generation of vacuum condition (pumping out with vent closed, steam-out then cooling) • Condensation • Gas dissolving in liquid • Restricted pump / compressor suction line • Undetected leakage • Vessel drainage • Power failure • Incorrect operation • Vacuum breaker or vent blocked or frozen • [S] Loss of furnace pressure control – air drawn into CO-rich or H2-rich system forming a flammable mixture • [S] Vacuum degassing (RH / VD / VOD) – vacuum pump or steam ejector failure, vessel seal leak • [S] Low fuel-gas pressure at burners (flame-out followed by re-ignition of accumulated gas) • [S] Low cooling-water pressure to water-cooled elements (pump trip, header leak)
Pressure	**Pressure fluctuation / surge**	• Water hammer; rapid valve closure; pump start/stop • Compressor or fan surge

Parameter	Deviation	Potential Causes
		• Slug flow / two-phase flow • Control loop instability • [S] Converter slopping or foaming slag eruption • [S] Explosive spalling of wet refractory
Temperature	**More temperature**	• Ambient conditions (hot weather, solar gain) • Fouled or failed exchanger tubes • Fire situation • Cooling water failure • Defective control (TIC fault, heater fails on) • Reaction control failures • Power failure (loss of cooling fans or pumps) • Heat tracing or jacket heating fails on • [C] Exothermic reaction / catalyst hot-spot; loss of quench • [S] Loss of cooling water to panels, roof, tuyeres, staves, lances or moulds – burn-through • [S] Refractory wear or loss – shell hot-spot, breakout • [S] Excessive metal superheat (power-on time too long, temperature measurement error) • [S] Hot DRI / HBI self-heating and re-oxidation in bins, silos or conveyors • [S] Reheat furnace over-firing; burner management failure • [S] Hot material on conveyors not rated for the temperature (belt fire)
Temperature	**Less temperature**	• Ambient conditions (cold weather, freezing) • Reducing pressure (auto-refrigeration of LPG, CO2, ammonia) • Fouled or failed exchanger tubes • Loss of heating (steam, hot oil, electric tracing) • Power failure • Brittle fracture of carbon steel below its minimum design metal temperature • [C] Crystallisation, hydrate formation or high viscosity • [S] Metal freeze in ladle, tundish, launder or nozzle (low tap temperature, delays) • [S] Insufficient pre-heat of ladles, tundishes, moulds or tools – moisture and thermal shock • [S] Low reducing-gas temperature – incomplete reduction / carbon deposition
Level	**High level**	• Inlet flow greater than outlet; outlet blocked or valve closed • Level controller or level transmitter failure • Wrong vessel being filled • Overfill during transfer or tanker offloading • [C] Foaming • [S] Over-filling of ladle at tap (molten metal / slag spill) • [S] High mould level in continuous caster (overflow)

Parameter	Deviation	Potential Causes
		• [S] Foaming slag over the sill / door • [S] Bin, silo or hopper over-fill (DRI, pellets, coal, lime) – spillage, structural overload
Level	**Low level / no level**	• Loss of feed; drain left open; leak • Level indication fails high (false reading) • Pump cavitation or loss of suction • Gas blow-by to downstream low-pressure vessel • [S] Low tundish level – slag entrainment, vortexing • [S] Low water level in seal pots or water seals – gas escape • [S] Low level in boiler drum or evaporative cooling system – overheating • [S] Empty bin or hopper allowing gas to pass back through feed system (loss of material seal)
Composition / Contamination	**Contamination / composition change (As well as)**	• Leaking exchanger tubes or isolation valves • Incorrect operation of system • Interconnected systems • Effect of corrosion • Wrong additives • Ingress of air, water or lubricant • Shutdown and start-up conditions • Phase change • Sediments / scale • Biological activity • Reaction by-products / breakdown products • [C] Catalyst poisons in feed • [S] Wet or contaminated scrap – sealed containers, explosives, oil, paint, plastics, radioactive sources • [S] Moisture in ferro-alloys, fluxes, ladle additions or tundish powders • [S] Oxygen or air ingress into reducing gas or inerted systems • [S] Hydrogen, CO or H2S in process gas above design values
Composition / Contamination	**Part of (wrong ratio / concentration)**	• Feed ratio controller failure; one feed lost • Dosing pump stopped or air-locked • Analyser or sample error • [C] Wrong acid or caustic concentration • [S] Carbon, sulphur or metallisation of DRI outside specification • [S] Wrong fuel / air ratio – incomplete combustion, CO formation • [S] Pickling acid strength out of range
Composition / Contamination	**Other than (wrong material)**	• Wrong raw material delivered or unloaded • Mislabelled drums, IBCs or cylinders • Wrong gas connected (e.g. oxygen in place of nitrogen, nitrogen in place of breathing air)

Parameter	Deviation	Potential Causes
		• Incorrect hose / coupling standardisation • [S] Wrong alloy addition or wrong grade of scrap charged
Phase	**Unintended phase change**	• Flashing, boiling or condensation • Freezing or solidification • Two-phase flow in single-phase design • [C] Vapour lock; liquid in compressor suction • [S] Molten metal–water contact – steam explosion • [S] Solidification of slag or metal in transfer equipment
Reaction	**More reaction / runaway**	• Loss of cooling or agitation • Excess catalyst or initiator; wrong reactant order • Higher feed concentration or temperature • Accumulation of unreacted material followed by sudden reaction • [C] Contamination promoting decomposition • [S] Excessive oxygen blowing – violent carbon boil / slopping • [S] Metal–water reaction generating hydrogen • [S] Aluminium, magnesium or other reactive metal powders in contact with water
Reaction	**Less / no reaction**	• Low temperature; catalyst deactivated • Wrong or missing reactant; inhibitor excess • [C] Accumulation of unreacted material (later runaway) • [S] Incomplete reduction in DRI shaft / kiln (low metallisation) • [S] Low reducing-gas quality or temperature
Reaction	**Unwanted side reaction (Other than)**	• [C] Decomposition, polymerisation, oxidation • [S] Carbon deposition (Boudouard reaction) in reducing-gas systems • [S] Carbonyl formation where CO contacts nickel or iron surfaces at low temperature • [S] Re-oxidation of DRI / HBI (heat and hydrogen release) • Toxic gas evolution (H2S, HCN, NOx, phosphine from wetted ferro-silicon)
Mixing / Agitation	**No / less / more mixing**	• Agitator, mixer or circulation pump trip • Wrong agitator speed • [C] Stratification of reactants • [S] Argon / nitrogen stirring – porous plug blocked or leaking • [S] Excessive stirring – metal splash, slag entrainment • [S] Electromagnetic stirrer failure
Time / Sequence	**Early / late / before / after / step omitted**	• Procedure step skipped or done out of order • Batch timer or sequence controller failure • Purge time too short before light-up (burner management)

Parameter	Deviation	Potential Causes
		• Operator distraction or shift changeover • [C] Reagent added before reaction mass at temperature • [S] Tapping late (over-oxidation, high temperature) or early (low temperature) • [S] Oxygen on before lance in correct position; lance not withdrawn at end of blow • [S] Charging scrap into a furnace with water present or hot heel too large • [S] Ladle not dried out / pre-heated before use
Time / Sequence	**Faster / slower**	• Ramp rate too fast for equipment or refractory • [S] Casting speed too fast – breakout • [S] Heat-up / dry-out of new refractory too fast – spalling • [S] Crane travel or tilting too fast – metal splash
Viscosity / Density	**More / less viscosity or density**	• Temperature change • Wrong material or concentration • Solids content too high • [S] Slag basicity or chemistry wrong – viscous slag, poor fluidity
Part B – Steel & Metals Industry Specific Parameters *Additional parameters for iron-making, DRI, steel-making (EAF / BOF / ladle metallurgy), casting, rolling, finishing and non-ferrous metals. Use alongside Part A.*		
Molten metal / slag containment	**Loss of containment (breakout, spill, splash)**	• Refractory wear, erosion or spalling; missed refractory inspection • Furnace shell or ladle shell burn-through • Tap-hole or slide-gate failure; nozzle failure • Ladle over-filling • Crane failure or dropped ladle; ladle trunnion failure • Ladle or pot not correctly seated on car or stand • Tilting drive over-travel; loss of hydraulic or electrical control • Breakout at caster (mould level, casting speed, lubrication, sticker) • Vehicle or ladle-car collision; derailment
Molten metal / slag containment	**Molten metal / slag contact with water (As well as)**	• Leaking water-cooled panel, roof, tuyere, stave, lance or copper mould • Wet scrap, DRI, fluxes or ladle additions; snow or ice • Wet or damp ladles, tundishes, moulds, tools or slag pots • Water on floor or in pits under tapping areas • Rain ingress through damaged roofs; flooding • Spray cooling water impinging on spilled metal • Granulation / slag-pit water management failure
Water-cooled elements	**No / less cooling water**	• Pump trip or power failure without emergency (gravity / diesel) supply • Valve closed in error; header isolation • Scale, silt or biological fouling

Parameter	Deviation	Potential Causes
		• Air locking of circuits • Low header pressure or tank level • Cooling tower failure (fans, basin level)
Water-cooled elements	**Water leak into furnace / vessel**	• Thermal fatigue cracking of panels or lance • Arc or flame impingement damage • Mechanical damage from scrap charging • Erosion or corrosion (water chemistry) • Poor weld or repair quality • Over-pressure of closed circuit • Leak detection (flow in/out differential) failed or not fitted
Process gas (CO, H2 – BFG, COG, LD gas, DRI reducing / top gas)	**Toxic release (Other than / As well as)**	• Flange, valve or seal leak; water-seal blown • Gas back-flow into buildings via drains or cable ducts • Bleeder / relief opening near work areas • Isolation failure during maintenance (no double block and bleed, no blinding) • Gas detection inoperative or poorly located • Confined spaces near gas mains
Process gas (CO, H2 – BFG, COG, LD gas, DRI reducing / top gas)	**Explosive mixture (air or oxygen in gas)**	• Air ingress through leaks at negative pressure • Inadequate purging at start-up / shutdown • Gas holder or collector low pressure • Oxygen analyser failure on LD gas recovery • Burner flame-out with continued fuel supply
Oxygen (lances, burners, oxygen pipelines)	**Oxygen fire / oxygen enrichment**	• Oil, grease or debris contamination in oxygen service • High gas velocity / particle impingement in valves and bends • Incompatible materials of construction or gaskets • Fast-opening valves (adiabatic compression) • Leak into enclosed areas – enriched atmosphere, clothing fire • Lance blocked or burned back
Inert gas (N2, Ar)	**Asphyxiation / loss of inerting**	• Leak or purge into enclosed or confined area • Inert gas used as breathing air (wrong connection) • Pit, tunnel or bin atmosphere not tested before entry • Loss of inert blanketing on DRI bins, coal mills or dust filters • Nitrogen supply failure (plant trip)
Electrical energy (EAF / LF / submerged arc / electrolysis)	**More / less / other than intended**	• Electrode breakage or slipping • Transformer or reactor cooling failure; transformer fire • Over-current or short circuit; regulation fault

Parameter	Deviation	Potential Causes
		• Arc flash on high-current bus bars • Loss of furnace power at critical stage (metal freeze) • [S] Electrolysis cells / potlines – DC magnetic fields, anode effect, cell tapping errors
Solid charge materials (scrap, DRI / HBI, pellets, coke, coal, fluxes, ferro-alloys)	**Wrong / contaminated / excess charge**	• Sealed containers, gas cylinders, explosives in scrap • Radioactive sources in scrap (portal monitor failure) • Moisture in charge • Oversize pieces – bridging, mechanical damage • Segregation – inconsistent chemistry • Fines – dust generation
Hot DRI / HBI handling	**Self-heating / fire / gas generation**	• Re-oxidation in air (insufficient inerting or cooling) • Water contact – heat and hydrogen generation • Dust and fines accumulation • Conveyor or transport vessel over-temperature • Long storage times; poor bin ventilation
Combustible dust (coal / PCI, coke, DRI fines, Al / Mg / Ti powders, carbon)	**Dust cloud / ignition**	• Dust accumulation on structures; poor housekeeping • Dust collector or baghouse fire / explosion • Hot surfaces, sparks, static, hot work • Loss of inerting in coal mill or PCI system • Explosion venting / suppression / isolation missing or ineffective
Hydraulic and lubrication oil systems (mills, casters, furnace tilt, cranes)	**Leak / fire / loss of pressure**	• Hose or fitting failure near hot metal or hot product • Use of mineral oil instead of fire-resistant fluid near hot areas • Accumulator failure – uncontrolled movement • Loss of pressure – uncontrolled tilt, roll gap or clamp release • Contaminated oil – valve sticking
Mechanical handling (cranes, ladle cars, transfer cars)	**Failure / collision / dropped load**	• Crane brake, rope or hook failure while carrying molten metal • Power failure with ladle suspended • Anti-collision or limit switch failure • Overload; incorrect lifting accessory • Poor visibility from steam or fume
Casting and rolling	**Casting speed / mould level / product handling deviations**	• Mould level control failure – overflow or breakout • Secondary cooling water failure • Cobbles in rolling mill; product jumping guides • Roll breakage; drive failure • Scale-breaker high-pressure water hazards

Parameter	Deviation	Potential Causes
Refractory	**Less / failed refractory**	• Wear beyond limit; inspection missed • Wrong refractory specification • Inadequate dry-out (moisture – spalling, steam) • Thermal shock from rapid heating or cooling • Chemical attack by slag
Acids, pickling, coating and galvanising	**Concentration / temperature / contamination deviations**	• Acid concentration out of range – fume, over-pickling • Acid regeneration plant upset – HCl release • Fume extraction failure • Wet or contaminated steel entering molten zinc bath – eruption • Flux or zinc ash handling exposures
Part C – Auxiliary Guidewords for Each Line / Node *From the original draft, applied to each line after the process parameters. Deviations are phrased as what can go wrong.*		
Relief	**Inadequate / inappropriate relief**	• Relief philosophy not defined for all scenarios (fire, blocked outlet, runaway, tube rupture) • Type of relief device and its reliability • Relief valve discharge location unsafe (to work areas, ignition sources) • Pollution implications of discharge • Relief isolated or blocked; incorrect set pressure • **[S]** Explosion doors / bleeders on furnaces and gas mains – location and maintenance
Process control / instrumentation	**Control or safety instrumented function failure**	• Control philosophy unclear • Location of instruments unsuitable • Response time; set points of alarms and trips • Time available for operator intervention • Alarm and trip testing not done • Fire protection of instruments and cables • Electronic trip / control amplifiers • Panel arrangement and location • Auto / manual facility and human error • Alarm flooding; bypassed trips • **[S]** Instruments in high temperature, dust and EMF environments (temperature probes, level radars, leak detection)
Sampling	**Sampling error / exposure**	• Sampling procedure inadequate • Time for analysis result too long • Calibration of automatic samplers / reliability • Accuracy of representative sample • Diagnosis of result • Exposure of sampler to toxic, hot or pressurised material

Parameter	Deviation	Potential Causes
		• [S] Molten metal sampling and temperature measurement at the furnace door or ladle
Corrosion / erosion	**More corrosion / erosion**	• Cathodic protection arrangements inadequate • Internal / external corrosion protection • Engineering specifications • Zinc embrittlement • Stress corrosion cracking (chlorides) • Fluid velocities too high • Corrosion under insulation • [S] Abrasive slurries, dust-laden gas, acid dew-point corrosion in off-gas ducts
Service failure	**Loss of utility / service**	• Failure of dry instrument air • Failure of steam / nitrogen / cooling water • Failure of hydraulic power / electric power • Telecommunications • Heating and ventilating systems • Computers / control system / network • [S] Loss of oxygen, argon or natural gas supply from air-separation plant or pipeline
Maintenance	**Unsafe maintenance conditions**	• Isolation, drainage, purging • Cleaning, drying • Slip plates • Access • Rescue plan • Training • Pressure testing • Work permit system • Condition monitoring • [S] Hot work near gas mains or dust; work under suspended loads; work inside cooled-down furnaces
Spare equipment	**Unavailability / wrong spares**	• Installed / non-installed spare equipment • Availability of spares • Modified specifications • Storage of spares • Catalogue of spares
Electrical	**Ignition source / electrical hazard**	• Static discharge • Earthing arrangements • Insulated vessels / equipment • Low-conductance fluids

Parameter	Deviation	Potential Causes
		• Splash filling of vessels • Insulated strainers and valve components • Dust generation and handling • Hoses • Area classification • **[S]** High-current bus bars, induced currents and magnetic fields
Human factors	**Operator error / omission**	• Unclear or absent procedures • Poor labelling and identification • Workload, fatigue, shift handover • Alarm overload; poor HMI design • Inadequate training or competence
Part D – Overview Guidewords (to be done when all lines have been studied) *Whole-system questions from the original draft. The Deviation column states the concern; the Potential Causes column lists what to review.*		
Simplicity	**Unnecessary complexity**	• Can we omit plant or simplify control? • Too many interlocks or bypasses
Output – reliability and bottlenecks	**Less output / unreliability**	• Least reliable components • Most limiting components
Efficiency – losses	**Loss of material, energy or conversion**	• Where will we lose material, conversion, yield or energy?
Construction	**Construction hazards and impacts**	• Demolition • Impacts on existing site, existing operations and environment • Size of equipment • Special hazards – chemicals / dangerous goods, to workers, structural • Personal protective equipment
Commissioning	**Commissioning not controlled**	• Broadly how it will be done • Sequence • What difficulties and hazards can be foreseen • Review of procedures • Signage / tags / locks • Responsibilities • **[S]** Refractory dry-out and first heat
Start-up	**Unsafe start-up**	• First start-up methods, difficulties, hazards • Routine start-up methods, difficulties, hazards • Philosophy of control (auto, manual, local)

Parameter	Deviation	Potential Causes
		• Too many interlocks? • Procedures • Purging and gas-freeing before light-up
Shutdown (isolation, purging)	**Unsafe shutdown**	• Planned shutdown • Emergency shutdown • Difficulties, hazards • [S] Holding molten metal during an extended stop; emptying furnaces and ladles
Breakdown (including service failure)	**Breakdown / loss of services**	• Power failure • Air, steam, water, vacuum, fuel • Vents • Computer • Other • Major causes of breakdown • Emergency procedures and communications • Which way should valves move (fail-safe positions)?
Materials	**Wrong / inadequate materials of construction**	• Vessels • Pipelines • Pumps • Valves • Gaskets / seals • Protective coatings • Others (refractory, copper, hoses)
Utilities and services	**Inadequate utilities**	• Lighting • Compressed air; instrument air • Nitrogen • Water (cooling, process, service, potable) • Steam • Power • Oxygen
Toxicity	**Toxic exposure**	• Material safety data sheets • Handling procedures • Precautions • Exposure monitoring • Emergency procedures • Ventilation

Parameter	Deviation	Potential Causes
		• [S] CO, metal fume, silica, PAHs from coke and pitch
Fire and explosion	**Fire / explosion**	• Broad overview • Protection, prevention, control • Location of safety showers, deluges • Location of shut-off valves and controls • [S] Molten metal explosions, dust explosions, gas explosions, hydraulic oil fires
Environmental impact	**Release to environment**	• Vapour emissions • Dust emissions • Effluent • Noise • Ground seepage • Waste minimisation • [S] Slag, mill scale, sludge, EAF dust disposal
Natural hazards	**External natural event**	• Earthquake • Flooding • Thunderstorm (lightning protection) • High winds • Sandstorms; extreme heat
Quality and consistency	**Off-spec product / variability**	• What will most cause quality to fail or vary? • How to prevent it
Procedures – development and documentation	**Missing / inadequate procedures**	• Operations • Maintenance • Confined space • Inspection and testing • Emergency • Engineering drawings
Safety equipment	**Missing / inadequate safety equipment**	• Personal protective equipment (PPE) • Fire detection • Fire fighting • Means of escape • Gas monitors • Breathing apparatus • Safety shower • Barriers / guards • [S] Heat-protective clothing; molten-metal splash protection

Parameter	Deviation	Potential Causes
Egress / access	**Inadequate egress or access**	• Number of exit points • Emergency egress • Size – width, height, length • Obstructions • Lighting • Fire protection
Entry / exit points	**Unsafe entry / exit arrangement**	• Location • Arrangement • Communications • External impacts
Layout and facility siting	**Domino / escalation effects**	• Spacing between hazardous units • Occupied buildings near hazards • Vehicle routes near molten metal or gas mains • Simultaneous operations (SIMOPS)